

Java Dojo – Detailed Plan with Examples & Exercises

Phase 1 — Functional & Modern Java

Example: Convert a list of orders into total revenue using streams.

Exercise: Refactor nested loops into stream pipelines.

Exercise: Implement a custom collector.

Phase 2 — Collections & Performance

Example: Compare ArrayList vs LinkedList for insert operations.

Exercise: Benchmark HashMap vs ConcurrentHashMap.

Exercise: Analyze memory usage of objects.

Phase 3 — Concurrency Fundamentals

Example: Shared counter with race condition.

Exercise: Fix using synchronized and AtomicInteger.

Exercise: Implement producer-consumer with wait/notify.

Phase 4 — Advanced Concurrency

Example: Parallel task execution with ExecutorService.

Exercise: Build a task scheduler.

Exercise: Compare ForkJoin vs single-thread execution.

Phase 5 — Async & Reactive

Example: Chain API calls using CompletableFuture.

Exercise: Add timeout, retry, fallback.

Exercise: Convert blocking code to async.

Phase 6 — Clean Code & Architecture

Example: Refactor tightly coupled service into interfaces.

Exercise: Apply SOLID principles.

Exercise: Introduce dependency injection manually.

Phase 7 — Design Patterns

Example: Strategy pattern for payment methods.

Exercise: Implement Observer for event system.

Exercise: Decorator for logging.

Phase 8 — Final Project

Example: Build async order processing system.

Exercise: Combine concurrency, async, and patterns.